Lever Test · Staff Answers

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Arrival Answers

1. Box 2 needed fewer coins (3 coins).
2. The pivot.
3. The load and its distance from the pivot.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Doubling the effort distance from the pivot halves the effort needed to balance and lift the load.

Hinge: A. Using the pattern, 15 cm lies between 10 cm and 20 cm, so the effort needed is between 4 coins and 2 coins (about 3 coins).

Evidence Desk Cards

Effort distance from the pivot (5 cm, 10 cm, 20 cm): Independent variable: we change the effort distance on purpose.

Number of identical coins needed to lift the load: Dependent variable: we measure the effort needed at each mark.

Mass of the load and its 5 cm distance from the pivot: Control variable: keeping the load and its position fixed makes the test fair.

At 20 cm it took 2 coins compared with 8 coins at 5 cm: Pattern & conclusion: increasing the effort distance reduces the effort needed.

Using identical 2p coins for every effort count: Control variable: identical coins make each unit of effort equal.

Predicting 3 coins at 15 cm from the pivot: Pattern & conclusion: using our results trend to make a sensible prediction.

Check & Repair Answer

Change only the effort distance; keep the load mass and its distance from the pivot the same.

Exit Ticket Answers

1. 20 cm (needed 2 coins compared with 8 coins at 5 cm). 2. The effort needed decreases. 3. Same load mass, same load distance from the pivot, same ruler/pivot, and identical coins.